The Target breach underscores the critical importance of not only knowing *who* is accessing your systems (authentication) but also controlling *what* they can access (authorization). The HVAC vendor had access to Target's systems for electronic billing, contract submission, and project management. However, the hackers were able to extend this access beyond those legitimate uses to reach systems used to update point-of-sale devices. Target's failure to properly restrict access to critical systems created a significant vulnerability that the hackers exploited.

But I don't want you to get the idea that authorization is just about securing IT systems—it's also fundamental to how modern cloud applications function, enabling features that wouldn't be possible otherwise. For example, services like Google Docs and Dropbox rely on authorization to allow users to share documents with specific people while restricting access to others. Without this capability, these applications would not be possible.

Another example is Amazon Web Services (AWS), where authorization enables Amazon to securely provide hundreds of cloud services to millions of customers at scale in a cost-effective manner. As a globally distributed system taking advantage of multi-tenancy to service millions of customers, AWS uses a sophisticated policy-based authorization system to help their customers control access by employees and others to the services running inside their accounts.

This book is about treating authorization as architecture, not just a checklist or security capability. It's written for architects, developers, identity teams, product managers, and technical leaders who are building or operating modern, distributed systems and need access control that scales with complexity while enabling product features safely and predictably. Over the next nineteen chapters, we'll move from foundational concepts to practical implementation, exploring dynamic authorization models, policy languages, governance practices, and emerging challenges like access control for AI agents. By the end, you'll be able to design, implement, and govern policy-based authorization systems that are flexible, auditable, and aligned with business intent, turning authorization from an afterthought into a strategic capability. To understand why authorization plays this central role, we have to start with something more fundamental: digital identity systems exist to manage relationships.